

The curvature of left-invariant magnetic systems

Shane Rankin, Ivo Terek, and David Weed

ABSTRACT. We compute the magnetic curvatures (in the sense of Assenza) of left-invariant magnetic systems on Lie groups and explore relations between curvature properties and algebraic properties, extending to the magnetic setting a number of results from Milnor's classic 1976 paper on the geometry of left-invariant metrics. We also discuss the existence of non-trivial bi-invariant magnetic systems and exhibit the resulting curvature formulas in some concrete examples, including the Heisenberg group.

1. Introduction

A *magnetic system* on a smooth manifold M is defined, as usual, to be a pair (g, σ) , where g is a Riemannian metric and σ is a closed 2-form on M ; in this context, σ is called the *magnetic 2-form*. One may then consider the second-order ordinary differential equation

$$(1.1) \quad \frac{D\dot{\gamma}}{dt}(t) = \mathbf{Y}_{\gamma(t)}(\dot{\gamma}(t)),$$

for smooth curves $\gamma: I \rightarrow M$, where the covariant derivative D/dt is induced by the Levi-Civita connection of g , and the *Lorentz force operator* $\mathbf{Y}: TM \rightarrow TM$, skew-adjoint and equivalent to σ under g , is characterized by

$$(1.2) \quad \sigma_x(v, w) = g_x(\mathbf{Y}_x(v), w), \quad \text{for all } x \in M \text{ and } v, w \in T_x M.$$

Equation (1.1) is sometimes called the *Landau-Hall equation*, and its solutions are called (g, σ) -geodesics; they all have constant speed due to skew-adjointness of $\mathbf{Y}$, and (g, σ) is called *complete* when all inextendible (g, σ) -geodesics are defined on all of $\mathbb{R}$. In this case, the magnetic flow obtained from (1.1) may be restricted to each energy level $\Sigma_s = \{(x, v) \in TM : \|v\|_x = s\}$, where $s > 0$. The dynamical and geometric properties of such magnetic flows, that now strongly depend on the

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value of s (unlike in the classical Riemannian case, where $\sigma = 0$), have been—and continue to be—largely investigated. See for instance [12, 19, 26, 35, 41] and the references therein, or [1, 2, 6, 15, 33] for more recent work.

With the recent introduction by Assenza [7] of a *magnetic curvature operator*, and the companion notions of *magnetic sectional and Ricci curvatures*, many results in Riemannian geometry have been generalized to the setting of magnetic systems. For example, we have the Bonnet-Myers and Synge’s theorems in [7], Green and Hopf’s theorems in [9], compatibility equations from submanifold theory have been listed in [39], and an interpretation of magnetic curvature as the curvature of a certain Cartan connection is given in the recent preprint [14]. We review the precise definitions of magnetic curvature, all of which explicitly carry the parameter $s > 0$ from the previous paragraph, in Section 2.

Here, we consider the simplest prototypes of homogeneous spaces: Lie groups. Naturally, a magnetic system on a Lie group is *left-invariant* if all left-translations are *magnetomorphisms* (that is, diffeomorphisms preserving both the metric and magnetic 2-form, cf. [2]). Explicit descriptions of magnetic geodesics have been obtained in Heisenberg groups [25, 37], and in three-dimensional groups with bi-invariant metrics [40], while integrability of magnetic flows (both in the Liouville and non-commutative senses) on Lie groups and homogeneous spaces has been thoroughly studied in [17, 23, 24, 32]. Our goal here is to study curvature. Following in spirit parts of the presentation by Milnor [36], we register formulas for the magnetic curvatures of left-invariant magnetic systems in Section 4, and discuss some relations between curvature properties and algebraic ones.

Our main results follow the general trend of many rigidity-related theorems for magnetic systems, where the magnetic 2-form ultimately vanishes; see e.g. [21, Lemma A.7], [8, Corollary 1.6], [34, Theorem A and Corollary 3.6], and [9, Theorem D]. The first one we generalize is a well-known consequence of [36, formula (7.3)], about the sectional curvature of a bi-invariant metric:

THEOREM A. *For any Lie group G and bi-invariant magnetic system (g, σ) on G , and any $s > 0$, it holds that $\text{Sec}^{(g, \sigma, s)} \geq 0$. Moreover, $\text{Sec}^{(g, \sigma, s)} = 0$ if and only if G is Abelian and $\sigma = 0$.*

As in [36], Theorem A follows from a more general computation—summarized in Proposition 4.3—of the s -magnetic sectional curvature along directions which are “compatible” with (g, σ) , in the sense of Proposition 4.1.

Next, we have a magnetic version of [36, Theorem 1.5], which specializes [9, Theorem D] to the left-invariant setting:

THEOREM B. *Let $(\mathfrak{g}, \sigma)$ be a left-invariant magnetic system on a connected Lie group G . Then, there is $s > 0$ such that $\text{Sec}^{(\mathfrak{g}, \sigma, s)} = 0$ if and only if $\nabla \sigma = 0$ and one of the following occurs:*

- (i) $\sigma = 0$ and the Lie algebra of G splits as an orthogonal direct sum $\mathfrak{g} = \mathfrak{b} \oplus \mathfrak{u}$, where $\mathfrak{b}$ is a commutative subalgebra, $\mathfrak{u}$ is a commutative ideal, and ad_b is skew-adjoint for every $b \in \mathfrak{b}$.
- (ii) σ is nowhere-vanishing and $J = \|\mathbf{Y}\|^{-1} \mathbf{Y}$ makes $(G, \mathfrak{g})$ a Kähler manifold holomorphically isometric to a complex hyperbolic space $\mathbb{C}\mathbb{H}^n$.

The main ingredient in the proof of Theorem B, perhaps hinted at by item (ii) therein, is a classical result by Heintze: every connected homogeneous Kähler manifold of negative curvature is holomorphically isometric to a complex hyperbolic space [27, Theorem 4].

Finally, we also obtain a magnetic version of [36, Theorem 1.6]:

THEOREM C. *Let G be a connected Lie group, and $(\mathfrak{g}, \sigma)$ be a left-invariant magnetic system on G . If there is $s > 0$ such that $\text{Sec}^{(\mathfrak{g}, \sigma, s)} = 0$, then G is solvable. If, in addition, G is unimodular, then $\sigma = 0$ and $\mathfrak{g}$ is flat.*

The main assumption in Theorem C is much more restrictive than the one in [36, Theorem 1.6], which allows the sectional curvature to be nonpositive. We conjecture, however, that Theorem C remains true if one instead assumes that $\text{Sec}^{(\mathfrak{g}, \sigma, s)} \leq 0$. The difficulties with a possible proof seem to ultimately stem from the general failure of the Hopf-Rinow theorem in the magnetic setting [13, Remark 2.5(b)], causing a crucial step in the proof of the corresponding would-be magnetic Cartan-Hadamard theorem to fail. Without it, it is not clear how to trace out the extensive work of Wolf [42], Azencott and Wilson [10, 11], Kobayashi [29], and others in order to obtain a classification of negatively curved homogeneous magnetic systems.

Organization of the paper. Throughout the entire text, unless otherwise stated, we work in the smooth category, G denotes a connected Lie group, and $\mathfrak{g}$ is its Lie algebra. In Sections 2 and 3 we review basic facts about magnetic curvature and left-invariant metrics. In Section 4 we put such facts together and register in Proposition 4.2 an explicit formula for the s -magnetic sectional curvature of left-invariant magnetic systems; a few consequences—including Theorem A—are then obtained. Next, in Section 5 we discuss the existence of nontrivial bi-invariant magnetic systems, and in Section 6 we prove Theorems B and C. Finally, in Section 7 we compute

the magnetic curvatures of a special class of systems on Heisenberg groups. Appendix A derives a generalization of the well-known Euler-Arnold formalism, for left-invariant exact magnetic systems.

2. Magnetic curvature

We briefly review, for the reader's convenience, the definitions of magnetic curvature presented by Assenza [7]. See also [9, pp. 2–3], [34, p. 2], and [39, p. 4].

Let (g, σ) be a magnetic system on a smooth manifold M , and let $UM \rightarrow M$ be the unit-tangent bundle of (M, g) . We consider two auxiliary bundles: the Stiefel bundle $\text{St}_2(M, g) \rightarrow M$ of orthonormal 2-frames tangent to M , and the bundle $E \rightarrow UM$ of orthogonal hyperplanes tangent to M ; specifically, their fibers are given by $\text{St}_2(M, g)_x = \{(v, w) \in (U_x M)^{\times 2} : g_x(v, w) = 0\}$ and $E_{(x,v)} = v^\perp$.

For each $s > 0$, the s -magnetic curvature operator of (g, σ) is the self-adjoint endomorphism $\mathbf{M}^{(g, \sigma, s)} : E \rightarrow E$ given by

$$(2.1) \quad \mathbf{M}_{(x,v)}^{(g, \sigma, s)}(w) = R_{(x,v)}^{(g, \sigma, s)}(w) + A_{(x,v)}^{(g, \sigma)}(w),$$

where $R^{(g, \sigma, s)}, A^{(g, \sigma)} : E \rightarrow E$ are in turn defined as

$$(2.2) \quad \begin{aligned} A_{(x,v)}^{(g, \sigma)}(w) &= -\frac{3}{4} \mathbf{Y}_x(\mathbf{Y}_x(w)^\top) - \frac{1}{4} \mathbf{Y}_x^2(w)^\perp, \quad \text{and} \\ R_{(x,v)}^{(g, \sigma, s)}(w) &= s^2 R_x^g(w, v)v - s(\nabla_w \mathbf{Y})(v) + \frac{s}{2}(\nabla_v \mathbf{Y})(w)^\perp, \end{aligned}$$

where R^g is the Riemann curvature tensor of g , and the tangent and perpendicular projections are taken relative to the orthogonal splitting $T_x M = \mathbb{R}v \oplus v^\perp$ induced by g . The general formulas

$$(2.3) \quad \begin{aligned} g_x(A_{(x,v)}^{(g, \sigma)}(w), w) &= \frac{3}{4} g_x(\mathbf{Y}_x(v), w)^2 + \frac{1}{4} \|\mathbf{Y}_x(w)\|^2 \\ \text{and } \text{tr}(A_{(x,v)}^{(g, \sigma)}) &= \frac{1}{2} \|\mathbf{Y}_x(v)\|^2 + \frac{1}{4} \|\mathbf{Y}_x\|^2, \quad \text{where } \|\mathbf{Y}_x\| = \sqrt{-\text{tr}(\mathbf{Y}_x^2)}, \end{aligned}$$

clear from (2.2), will be useful in Sections 4 and 7. The s -magnetic sectional curvature $\text{Sec}^{(g, \sigma, s)} : \text{St}_2(M, g) \rightarrow \mathbb{R}$ and s -magnetic Ricci curvature $\text{Ric}^{(g, \sigma, s)} : UM \rightarrow \mathbb{R}$ are defined by

$$(2.4) \quad \text{Sec}_x^{(g, \sigma, s)}(v, w) = g_x(\mathbf{M}_{(x,v)}^{(g, \sigma, s)}(w), w) \quad \text{and} \quad \text{Ric}^{(g, \sigma, s)}(x, v) = \text{tr}(\mathbf{M}_{(x,v)}^{(g, \sigma, s)}).$$

The s -magnetic sectional curvature does not survive as a function on the Grassmannian bundle of 2-planes tangent to M with,

$$(2.5) \quad \text{in general, } \text{Sec}_x^{(g, \sigma, s)}(v, w) \neq \text{Sec}_x^{(g, \sigma, s)}(w, v).$$

As in the Riemannian case, however, the s -magnetic Ricci curvature may be expressed in terms of the s -magnetic sectional curvature, with

$$(2.6) \quad \text{Ric}^{(\mathbf{g}, \sigma, s)}(x, v) = \sum_{i=1}^{n-1} \text{Sec}_x^{(\mathbf{g}, \sigma, s)}(v, e_i),$$

where $e_1, \dots, e_{n-1}$ is any orthonormal basis of $v^\perp$. Some useful relations between the magnetic curvatures and the Riemannian ones are also readily obtained, e.g.

$$(2.7) \quad \text{Sec}_x^{(\mathbf{g}, \sigma, s)}(v, w) = s^2 \text{Sec}_x^{\mathbf{g}}(v, w) - s g_x((\nabla_w \mathbf{Y})(v), w) + g_x((A_{(x,v)}^{(\mathbf{g}, \sigma)}(w), w),$$

cf. [9, Formula (6.3)], and from there with (2.6),

$$(2.8) \quad \text{Ric}^{(\mathbf{g}, \sigma, s)}(x, v) = s^2 \text{Ric}_x^{\mathbf{g}}(v, v) - s (\text{div}_{\mathbf{g}} \sigma)_x(v) + \text{tr}(A_{(x,v)}^{(\mathbf{g}, \sigma)}),$$

where $\text{div}_{\mathbf{g}} \sigma \in \Omega^1(M)$ is given by $(\text{div}_{\mathbf{g}} \sigma)(Y) = \text{tr}_{\mathbf{g}}((X, Z) \mapsto (\nabla_X \sigma)(Y, Z))$.

3. Riemannian curvature of Lie groups

We present, for later reference and to establish notation, some general facts about left-invariant metrics on Lie groups. In particular, we refer to [20, Proposition 3.18], [16, Theorem 7.30], or [5, Theorem 5.3] for more details on the formulas involving curvature.

Let G be a connected Lie group, $\mathfrak{g}$ be its Lie algebra, and $\mathbf{g} = \langle \cdot, \cdot \rangle$ be a left-invariant metric on G . For all $x, y \in \mathfrak{g}$, we set $\text{ad}_x y = [x, y]$, and denote by $\text{ad}_x^\dagger$ the $\mathbf{g}$ -adjoint of ad_x , characterized by $\langle \text{ad}_x y, z \rangle = \langle y, \text{ad}_x^\dagger z \rangle$ for all $z \in \mathfrak{g}$. The Levi-Civita connection of $\mathbf{g}$ is given by the Koszul formula

$$(3.1) \quad \nabla_x y = \frac{1}{2} [x, y] - B(x, y), \quad \text{where} \quad B(x, y) = \frac{1}{2} (\text{ad}_x^\dagger y + \text{ad}_y^\dagger x).$$

We also have that $\mathbf{g}$ is bi-invariant if and only if $B = 0$, as a consequence of the easily established formula $-2\langle B(x, y), z \rangle = \langle x, \text{ad}_z y \rangle + \langle \text{ad}_z x, y \rangle$. Indeed, $\mathbf{g}$ is bi-invariant if and only if each ad_z is a skew-adjoint operator [36, Lemma 7.2]. Moreover, it directly follows from (3.1) that, whenever $z, v \in \mathfrak{g}$ and ad_z is skew-adjoint, the relations

$$(3.2) \quad \text{(i) } \nabla_z z = 0, \quad \text{(ii) } \nabla_v z = -\frac{1}{2} \text{ad}_v^\dagger z, \quad \text{(iii) } \nabla_z v = \text{ad}_z v - \frac{1}{2} \text{ad}_v^\dagger z$$

all hold. Finally, we have that

$$(3.3) \quad \begin{aligned} \text{Sec}^{\mathbf{g}}(v, w) = & -\frac{3}{4} \|[v, w]\|^2 - \frac{1}{2} \langle [[v, w], w], v \rangle \\ & - \frac{1}{2} \langle [v, [v, w]], w \rangle - \langle B(v, v), B(w, w) \rangle + \|B(v, w)\|^2 \end{aligned}$$

whenever $v, w \in \mathfrak{g}$ are orthonormal vectors; in particular, whenever $B = 0$, (3.3) reduces to $\text{Sec}^{\mathbf{g}}(v, w) = \|[v, w]\|^2/4$.

4. Left-invariant magnetic systems and sectional curvature

Let G be a connected Lie group, and $\mathfrak{g}$ its Lie algebra. A left-invariant 2-form σ on G is closed if and only if

$$(4.1) \quad \sigma([u, v], w) + \sigma([v, w], u) + \sigma([w, u], v) = 0, \quad \text{for all } u, v, w \in \mathfrak{g}.$$

If σ is exact, and G is compact or has $H^2(\mathfrak{g}, \mathbb{R}) = \{0\}$, we may choose a left-invariant 1-form α on G such that $\sigma = -d\alpha$, in which case $\sigma(u, v) = \alpha([u, v])$ for all $u, v \in \mathfrak{g}$. If $g = \langle \cdot, \cdot \rangle$ is a left-invariant metric on G , we will often start with a skew-adjoint operator $\mathbf{Y}$ on $\mathfrak{g}$, and *define* σ via (1.2) instead; it follows from (1.2) and (4.1) that

$$(4.2) \quad \text{if } [\mathfrak{g}, \mathfrak{g}] \subseteq \ker \mathbf{Y}, \text{ the corresponding 2-form } \sigma \text{ is closed.}$$

Next, a left-invariant covariant tensor field Θ is bi-invariant if and only if it is Ad-invariant. This latter condition, as G is connected, may be tested for infinitesimally: it holds if and only if

$$(4.3) \quad \sum_{i=1}^k \Theta(v_1, \dots, v_{i-1}, \text{ad}_z v_i, v_{i+1}, \dots, v_k) = 0,$$

for all $z, v_1, \dots, v_k \in \mathfrak{g}$. It now follows that

$$(4.4) \quad \text{if the metric } g \text{ is bi-invariant, } \Theta \text{ is bi-invariant if and only if } \Theta \text{ is parallel.}$$

Indeed, the left side of (4.3) equals $-2(\nabla_z \Theta)(v_1, \dots, v_k)$, due to (3.1) and the definition of $\nabla_z \Theta$. Bi-invariant magnetic systems will be discussed in more detail in Section 5, but we may already register a very useful characterization which follows from the above discussion.

PROPOSITION 4.1. *A left-invariant magnetic system (g, σ) on a connected Lie group G is bi-invariant if and only if, for every $z \in \mathfrak{g}$, ad_z is skew-adjoint and commutes with $\mathbf{Y}$.*

PROOF. We know from [36, Lemma 7.2] that g is bi-invariant if and only if for each $z \in \mathfrak{g}$, ad_z is skew-adjoint. In this case (3.1) yields $2\nabla_z \mathbf{Y} = [\text{ad}_z, \mathbf{Y}]$ and, noting that σ is parallel if and only if $\mathbf{Y}$ is parallel, we may invoke (4.4). $\square$

We now proceed to compute the magnetic curvature of a left-invariant magnetic system (g, σ) . The curvatures will always be evaluated at vectors in $\mathfrak{g}$, and so we omit the identity base point $e \in G$, using the respective shorthand notations $\mathbf{Y}$, $A_v^{(g, \sigma)}$, $\text{Sec}^{(g, \sigma, s)}(v, w)$, and $\text{Ric}^{(g, \sigma, s)}(v)$ for $\mathbf{Y}_e$, $A_{(e, v)}^{(g, \sigma)}$, $\text{Sec}^{(g, \sigma, s)}(e, v)$, and

$\text{Ric}^{(\mathfrak{g}, \sigma, s)}(e, v)$, where $v, w \in \mathfrak{g}$. First, we claim that

$$(4.5) \quad \begin{aligned} \langle (\nabla_w \mathbf{Y})(v), w \rangle &= \langle [w, \mathbf{Y}(v)], w \rangle + \frac{1}{2} \langle [w, v], \mathbf{Y}(w) \rangle \\ &\quad - \frac{1}{2} \langle [v, \mathbf{Y}(w)], w \rangle - \frac{1}{2} \langle [w, \mathbf{Y}(w)], v \rangle \end{aligned}$$

holds, for any $v, w \in \mathfrak{g}$. Indeed, by definition of $\nabla_w \mathbf{Y}$ and skew-adjointness of $\mathbf{Y}$, we have the relation $\langle (\nabla_w \mathbf{Y})(v), w \rangle = \langle \nabla_w(\mathbf{Y}(v)), w \rangle + \langle \nabla_w v, \mathbf{Y}(w) \rangle$. Using (3.1) and that $[w, w] = 0$, it follows that $\langle \nabla_w(\mathbf{Y}(v)), w \rangle = \langle [w, \mathbf{Y}(v)], w \rangle$. A similar calculation again using (3.1) shows that the second term $\langle \nabla_w v, \mathbf{Y}(w) \rangle$ amounts to the remaining three terms in (4.5). We register next the result of plugging (2.3), (3.3), and (4.5) into (2.7).

PROPOSITION 4.2. *Let $(\mathfrak{g}, \sigma)$ be any left-invariant magnetic system on G , and $s > 0$. Then, we have that*

$$(4.6) \quad \begin{aligned} \text{Sec}^{(\mathfrak{g}, \sigma, s)}(v, w) &= s^2 \left(-\frac{3}{4} \|[v, w]\|^2 - \frac{1}{2} \langle [[v, w], w], v \rangle - \frac{1}{2} \langle [v, [v, w]], w \rangle \right. \\ &\quad \left. - \langle B(v, v), B(w, w) \rangle + \|B(v, w)\|^2 \right) - s \left(\langle [w, \mathbf{Y}(v)], w \rangle \right. \\ &\quad \left. + \frac{1}{2} \langle [w, v], \mathbf{Y}(w) \rangle - \frac{1}{2} \langle [v, \mathbf{Y}(w)], w \rangle - \frac{1}{2} \langle [w, \mathbf{Y}(w)], v \rangle \right) \\ &\quad + \frac{3}{4} \langle w, \mathbf{Y}(v) \rangle^2 + \frac{1}{4} \|\mathbf{Y}(w)\|^2, \end{aligned}$$

whenever $v, w \in \mathfrak{g}$ are orthonormal, for B as in (3.1).

The next result generalizes [36, Lemma 1.6], which states that the Riemannian sectional curvature of a left-invariant metric is nonnegative along planes containing directions z with skew-adjoint ad_z . The new assumption considered below, which means that the one-parameter subgroup generated by z acts on G via conjugations preserving *both* $\mathfrak{g}$ and σ , is of course motivated by Proposition 4.1.

PROPOSITION 4.3. *Let $z \in \mathfrak{g}$ be a unit vector such that ad_z is skew-adjoint and commutes with $\mathbf{Y}$. Then, $\text{Sec}^{(\mathfrak{g}, \sigma, s)}(v, z) \geq 0$ for every $s > 0$ and unit vector $v \in \mathfrak{g}$ orthogonal to z , with equality if and only if z is orthogonal to the image of $\text{ad}_v - s^{-1}\mathbf{Y}$. In particular, this holds if z is a central element.*

PROOF. Using that z is orthogonal to the images of ad_z and $v \mapsto \text{ad}_v^\dagger z$ whenever ad_z is skew-adjoint, a routine computation reduces (3.3) to $\text{Sec}^{\mathfrak{g}}(v, z) = \|\text{ad}_v^\dagger z\|^2/4$. Next, using also the condition that $[\text{ad}_z, \mathbf{Y}] = 0$, only the third term in (4.5) remains, which we write as $\langle (\nabla_z \mathbf{Y})(v), z \rangle = -\langle \mathbf{Y}(z), \text{ad}_v^\dagger z \rangle/2$. Substituting all of it into (2.7) and factoring it with (2.3), we arrive at

$$(4.7) \quad \text{Sec}^{(\mathfrak{g}, \sigma, s)}(v, z) = \frac{s^2}{4} \|\text{ad}_v^\dagger z + s^{-1}\mathbf{Y}(z)\|^2 + \frac{3}{4} \langle \mathbf{Y}(z), v \rangle^2 \geq 0,$$

with equality if and only if $\text{ad}_v^\dagger z + s^{-1}\mathbf{Y}(z) = 0$ and $\mathbf{Y}(z)$ is orthogonal to v . However, the latter condition is implied by the former, and the conclusion follows from the general relation $\langle \text{ad}_v^\dagger z + s^{-1}\mathbf{Y}(z), w \rangle = \langle z, \text{ad}_v w - s^{-1}\mathbf{Y}(w) \rangle$, valid for all $w \in \mathfrak{g}$. $\square$

Theorem A now follows from Proposition 4.3, in view of Proposition 4.1 together with the easily verified claim that $\text{ad}_v^\dagger z + s^{-1}\mathbf{Y}(z)$ vanishes for all $v, z \in \mathfrak{g}$ if and only if $\text{ad}_v = 0$ for all v (i.e., G is Abelian) and $\mathbf{Y} = 0$ (hence $\sigma = 0$). Alternatively, one may simplify (4.6) in the case where $(\mathfrak{g}, \sigma)$ is bi-invariant to obtain

$$(4.8) \quad \text{Sec}^{(\mathfrak{g}, \sigma, s)}(v, w) = \frac{1}{4} (s^2 \| [v, w] \|^2 + 3 \langle w, \mathbf{Y}(v) \rangle^2 + \|\mathbf{Y}(w)\|^2),$$

for all orthonormal $v, w \in \mathfrak{g}$.

EXAMPLE 4.4. The assumption that ad_z commutes with $\mathbf{Y}$ in Proposition 4.3 cannot be dropped. Consider the Lie group $E(2)$ of all rigid motions of the plane, equipped with the left-invariant metric $\mathfrak{g}$ for which the basis x, y, z of $\mathfrak{e}(2)$ satisfying the bracket relations $[z, x] = -y$, $[z, y] = x$, and $[x, y] = 0$ declared to be $\mathfrak{g}$ -orthonormal. Note that every left-invariant 2-form on $E(2)$ is automatically closed: as $E(2)$ is three-dimensional, it suffices to check that (4.1) is satisfied when evaluated at $(u, v, w) = (x, y, z)$; this is obvious from the given bracket relations. The operator $\mathbf{Y}$ defined by $\mathbf{Y}(x) = 0$, $\mathbf{Y}(y) = z$, $\mathbf{Y}(z) = -y$, is a rotation in the yz -plane, and thus skew-adjoint; define σ via (1.2). The operator ad_z is also skew-adjoint, being another rotation, but $[\text{ad}_z, \mathbf{Y}](x) = z \neq 0$. Using (3.2), we compute that $\nabla_z x = -y$ and $\nabla_x z = \nabla_y z = \nabla_z z = 0$, so that $R^{\mathfrak{g}}(x, z)z = 0$ and $\langle (\nabla_z \mathbf{Y})(x), z \rangle = 1$. From (2.7), we finally obtain that $\text{Sec}^{(\mathfrak{g}, \sigma, s)}(x, z) = -s + 1/4 < 0$ for all $s > 1/4$.

EXAMPLE 4.5. In view of (2.5), one may also ask if the natural analogue of Proposition 4.3 holds, with $\text{Sec}^{(\mathfrak{g}, \sigma, s)}(z, w) \geq 0$ for all $s > 0$ and unit vector $w \in \mathfrak{g}$ orthogonal to z , provided that ad_z is skew-adjoint and commutes with $\mathbf{Y}$. This is, however, false: consider the group $\text{Aff}(\mathbb{C})$ of affine functions on $\mathbb{C}$, equipped with the left-invariant metric $\mathfrak{g}$ for which the basis x, y, z, w of $\mathfrak{aff}(\mathbb{C})$ having the nonzero bracket relations $[x, w] = w$, $[x, y] = y$, $[z, w] = y$, and $[z, y] = -w$, is $\mathfrak{g}$ -orthonormal. Using (3.1) and (3.2), we fully describe the Levi-Civita connection of $\mathfrak{g}$ in Table 1.

(4.9)

∇	x	y	z	w
x	0	0	0	0
y	$-y$	x	0	0
z	0	$-w$	0	y
w	$-w$	0	0	x

TABLE 1. The Levi-Civita product of $\mathfrak{g}$ in $\mathfrak{aff}(\mathbb{C})$.

The operator $\mathbf{Y}$ defined by $\mathbf{Y}(x) = z$, $\mathbf{Y}(y) = 0$, $\mathbf{Y}(z) = -x$, and $\mathbf{Y}(w) = 0$, being a rotation in the xz -plane, is skew-adjoint and defines a closed 2-form σ , by (1.2) and (4.2). It also commutes with ad_z , which is a similar rotation in the yw -plane, and hence skew-adjoint as well. Using (2.7) and Table 1, we find that

$$(4.10) \quad \text{Sec}^{(\mathfrak{g}, \sigma, s)}(z, x) = 1, \quad \text{and} \quad \text{Sec}^{(\mathfrak{g}, \sigma, s)}(z, y) = \text{Sec}^{(\mathfrak{g}, \sigma, s)}(z, w) = -s,$$

for all $s > 0$, making the latter expression manifestly negative.

REMARK 4.6. In [36, Lemmas 2.1 and 2.3], two results on Ricci curvature are presented: (i) If $z \in \mathfrak{g}$ has skew-adjoint ad_z , then $\text{Ric}^{\mathfrak{g}}(z) \geq 0$, with equality if and only if $z \in [\mathfrak{g}, \mathfrak{g}]^{\perp}$, and (ii) if $z \in [\mathfrak{g}, \mathfrak{g}]^{\perp}$, then $\text{Ric}^{\mathfrak{g}}(z) \leq 0$, with equality if and only if ad_z is skew-adjoint. Both are false in the magnetic setting. For (i), we recycle Example 4.5: it follows from (2.6) and (4.10) that $\text{Ric}^{(\mathfrak{g}, \sigma, s)}(z) = 1 - 2s < 0$ for $s > 1/2$. As for (ii), note from (2.3) that $\text{tr}(A_{(x,v)}^{(\mathfrak{g}, \sigma)}) \geq 0$, with equality if and only if $\mathbf{Y}_x = 0$, so that making $s \rightarrow 0^+$ in (2.8) shows that $\text{Ric}^{(\mathfrak{g}, \sigma, s)}(x, v) > 0$ for s sufficiently small whenever $\mathbf{Y}_x \neq 0$, regardless of any properties one might require of v .

5. More on the bi-invariant case

It is well-known that a Lie group admits a bi-invariant metric if and only if G is isomorphic to the product of a compact group and an Abelian group [36, p. 297]. We may clearly replace “bi-invariant metric” with “bi-invariant magnetic system” here, by simply considering systems of the form $(\mathfrak{g}, 0)$. This raises the question of when is it possible to find a *nontrivial* bi-invariant magnetic system, that is, one with $\sigma \neq 0$.

For example, rearranging (4.3), we have that

$$(5.1) \quad \begin{aligned} &\text{a bilinear form } \Theta \text{ is bi-invariant if and only} \\ &\text{if } \Theta([x, y], z) = \Theta(x, [y, z]) \text{ for all } x, y, z \in \mathfrak{g}. \end{aligned}$$

With this, we claim that

$$(5.2) \quad \text{if } \mathfrak{g} = [\mathfrak{g}, \mathfrak{g}] \text{ (that is, } \mathfrak{g} \text{ is perfect), the only bi-invariant bilinear form on } \mathfrak{g} \text{ is the trivial one.}$$

Indeed, if Θ is bi-invariant, the trilinear form $(x, y, z) \mapsto \Theta([x, y], z)$ is at the same time skew-symmetric in the first pair of arguments, and symmetric in the second pair by (5.1), and hence vanishes, implying that $\Theta([\mathfrak{g}, \mathfrak{g}], \mathfrak{g}) = 0$, and thus $\Theta = 0$ as required. A similar argument shows that, in general, every bi-invariant 2-form is automatically closed: applying bi-invariance of σ to each term in the left side of (4.1) yields $d\sigma = -d\sigma$.

Below, recall that a Lie algebra $\mathfrak{g}$ is called *reductive* if there is a direct-sum decomposition

$$(5.3) \quad \mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus \bigoplus_{i=1}^k \mathfrak{g}_i, \quad \text{with each of } \mathfrak{g}_1, \dots, \mathfrak{g}_k \text{ simple,}$$

where $\mathfrak{z}(\mathfrak{g})$ is the center of $\mathfrak{g}$. This is a natural condition to consider, being automatically satisfied whenever G admits a bi-invariant metric. Namely, in this case (5.1) ensures that $\mathfrak{z}(\mathfrak{g}) = [\mathfrak{g}, \mathfrak{g}]^\perp$, yielding

$$(5.4) \quad \text{the orthogonal direct-sum splitting } \mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}],$$

with $[\mathfrak{g}, \mathfrak{g}]$ semisimple: by (5.4), any Abelian ideal $\mathfrak{a} \subseteq [\mathfrak{g}, \mathfrak{g}]$ must in fact commute with all of $\mathfrak{g}$, leading to $\mathfrak{a} \subseteq \mathfrak{z}(\mathfrak{g}) \cap [\mathfrak{g}, \mathfrak{g}] = \{0\}$.

PROPOSITION 5.1. *If $\mathfrak{g}$ is reductive, the restriction mapping*

$$(5.5) \quad (\wedge^2 \mathfrak{g}^*)^G \ni \sigma \mapsto \sigma|_{\mathfrak{z}(\mathfrak{g}) \times \mathfrak{z}(\mathfrak{g})} \in \wedge^2 \mathfrak{z}(\mathfrak{g})^*$$

is an isomorphism; here, $(\wedge^2 \mathfrak{g}^)^G$ denotes the space of bi-invariant 2-forms on $\mathfrak{g}$.*

PROOF. For a decomposition of $\mathfrak{g}$ as in (5.3), and any bi-invariant 2-form σ on G , we first claim that

$$(5.6) \quad \text{(i) } \sigma(\mathfrak{g}_i, \mathfrak{g}_j) = 0, \text{ and (ii) } \sigma(\mathfrak{g}_i, \mathfrak{z}(\mathfrak{g})) = 0, \text{ for all } i, j = 1, \dots, k.$$

If $i = j$, (5.6-i) follows from (5.2) as each $\mathfrak{g}_i$, being simple, is also perfect. Assuming that $i \neq j$ instead, let $\mathfrak{b}$ be either $\mathfrak{g}_j$ or $\mathfrak{z}(\mathfrak{g})$, take $x, y \in \mathfrak{g}_i$ and $z \in \mathfrak{b}$, and note that $\sigma([x, y], z) = \sigma(x, [y, z]) = 0$ as $[\mathfrak{g}_i, \mathfrak{b}] = 0$, so that $\sigma([\mathfrak{g}_i, \mathfrak{g}_i], \mathfrak{b}) = 0$, leading to both (5.6-i) and (5.6-ii) due to perfectness of $\mathfrak{g}_i$. Finally, condition (5.6) amounts to injectivity of (5.5), while its surjectivity is clear: given a 2-form on $\mathfrak{z}(\mathfrak{g})$, extend it by zero via (5.3) and (5.6) to a 2-form on $\mathfrak{g}$; bi-invariance of the resulting extension is immediate from (5.1), as all relevant brackets vanish. $\square$

We may now establish the main result of this section.

THEOREM 5.2. *A connected Lie group G admits a non-trivial bi-invariant magnetic system if and only if G is isomorphic to a direct product $K \times \mathbb{R}^m$, where $\mathbb{R}^m$ is Abelian and K is a compact Lie group with $\dim \mathfrak{z}(\mathfrak{k}) + m \geq 2$.*

PROOF. It follows from Proposition 5.1 that, in the presence of a bi-invariant metric, nonzero bi-invariant 2-forms exist if and only if $\dim \mathfrak{z}(\mathfrak{g}) \geq 2$; this is precisely the condition needed for $\dim (\wedge^2 \mathfrak{g}^*)^G > 0$. We now invoke [36, p. 297], stated in the beginning of this section, and note that $\dim \mathfrak{z}(\mathfrak{g}) = \dim \mathfrak{z}(\mathfrak{k}) + m$. $\square$

COROLLARY 5.3. *A connected and semisimple Lie group does not admit non-trivial bi-invariant magnetic systems.*

EXAMPLE 5.4. If g is a bi-invariant metric on G , we may consider $\mathbf{Y} = \text{ad}_\xi$ for some $\xi \in \mathfrak{g}$; the corresponding 2-form σ_ξ obtained via (1.2) is closed: (4.1) becomes the Jacobi identity. This example, at first natural, has a crucial “flaw”: if (g, σ_ξ) is bi-invariant, then $\sigma_\xi = 0$. Indeed, in this case Proposition 4.1 tells us that $0 = [\text{ad}_\xi, \text{ad}_x] = \text{ad}_{[\xi, x]}$ for all $x \in \mathfrak{g}$, so that $[\xi, x] \in \mathfrak{z}(\mathfrak{g}) \cap [\mathfrak{g}, \mathfrak{g}] = \{0\}$, cf. (5.4), and thus $\mathbf{Y} = \text{ad}_\xi = 0$. When g is bi-invariant and σ_ξ is not, from (2.7) and (2.6) we still obtain

$$(5.7) \quad \begin{aligned} \text{Sec}^{(g, \sigma_\xi, s)}(v, w) &= \frac{1}{4} \| [sv - \xi, w] \|^2 + \frac{3}{4} \langle [\xi, v], w \rangle^2 \\ \text{and } \text{Ric}^{(g, \sigma_\xi, s)}(v) &= -\frac{1}{4} \beta(sv - \xi, sv - \xi) + \frac{3}{4} \| [\xi, v] \|^2 \end{aligned}$$

for all orthonormal $v, w \in \mathfrak{g}$, where β is the Killing form of $\mathfrak{g}$.

6. Magnetic flatness of left-invariant systems

The meaning of magnetic flatness for a general magnetic system (g, σ) on a smooth manifold M is well-understood:

$$(6.1) \quad \begin{aligned} &\text{If } s > 0 \text{ is such that } \text{Sec}^{(g, \sigma, s)} = 0, \text{ then } \nabla \sigma = 0 \text{ and one of the} \\ &\text{following must hold: (i) } \sigma = 0 \text{ and } g \text{ is flat, or (ii) } \sigma \text{ is nowhere-} \\ &\text{vanishing and } J = \|\mathbf{Y}\|^{-1} \mathbf{Y} \text{ turns } (M, g) \text{ into a Kähler manifold} \\ &\text{of constant negative holomorphic sectional curvature } -\|\mathbf{Y}\|^2/s^2, \end{aligned}$$

cf. [9, Theorem D]. Here, $\|\mathbf{Y}\|$ denotes the operator norm of $\mathbf{Y}$. The goal of this section is to refine (6.1) in the case of left-invariant magnetic systems on Lie groups, by proving Theorems B and C. One general observation we will need is that if a Kähler manifold has constant holomorphic sectional curvature k , then its sectional curvature is bounded between $k/4$ and k , cf. [30, Propositions 7.3 and 7.4]; in particular, it follows that

$$(6.2) \quad \text{if } k < 0, \text{ both the sectional and Ricci curvatures are also strictly negative.}$$

In both Theorems B and C the situation where $\sigma = 0$ amounts to [36, Theorems 1.5 and 1.6], and so it suffices to address the case where σ is nowhere-vanishing.

PROOF OF THEOREM B. If σ is nowhere-vanishing, it follows from (6.1-ii) and (6.2) that g is negatively curved, allowing us to invoke the result by Heintze mentioned in the Introduction—it follows that (G, g) with the complex structure $J = \|\mathbf{Y}\|^{-1}\mathbf{Y}$ is holomorphically isometric to a $\mathbb{CH}^n$ [27, Theorem 4]. Conversely, if (G, g) equipped with $J = \|\mathbf{Y}\|^{-1}\mathbf{Y}$ is holomorphically isometric to a $\mathbb{CH}^n$ with constant holomorphic curvature $k < 0$, then (g, σ) is in particular a Kähler magnetic system (as $\|\mathbf{Y}\|$ is constant by left-invariance of g and $\mathbf{Y}$), with s -magnetic sectional curvature given by

$$(6.3) \quad \text{Sec}^{(g, \sigma, s)}(v, w) = \frac{s^2 k + \|\mathbf{Y}\|^2}{4} (1 + 3\langle v, Jw \rangle^2),$$

cf. [9, Formula (6.2)]; it vanishes identically for $s = \|\mathbf{Y}\|/\sqrt{-k}$. $\square$

PROOF OF THEOREM C. As in the previous proof, the sectional curvature is strictly negative. By [27, Remark in p. 27], the unique connected and simply connected Lie group $\tilde{G}$ having $\mathfrak{g}$ as its Lie algebra is solvable. Being connected, G is a quotient of $\tilde{G}$, and hence solvable as well. Finally, if $\sigma \neq 0$, G cannot be unimodular, as otherwise (6.2) contradicts a classical result by Dotti: no solvable unimodular Lie group admits a left-invariant Riemannian metric of strictly negative Ricci curvature [22, Corollary 3.3]. $\square$

7. The Heisenberg group

Let (V, Ω) be a symplectic vector space, and denote by $H(V, \Omega)$ its abstract Heisenberg group. Namely, $H(V, \Omega)$ is the Cartesian product $V \times \mathbb{R}$ equipped with the group operation $(x, a)(y, b) = (x + y, a + b + \Omega(x, y)/2)$, and its Lie algebra $\mathfrak{h}(V, \Omega) = V \times \mathbb{R}$ has the bracket $[(v, a), (w, b)] = (0, \Omega(v, w))$. The group $H(V, \Omega)$ is two-step nilpotent, its center equals $\{0\} \times \mathbb{R}$ due to nondegeneracy of Ω , and so the resulting short exact sequence $0 \rightarrow \mathbb{R} \rightarrow H(V, \Omega) \rightarrow V \rightarrow 0$ yields that $H(V, \Omega)$ is a central extension of V by $\mathbb{R}$. Let $\langle \cdot, \cdot \rangle_0$ be any inner product on V , and consider the left-invariant magnetic system (g, σ) on $H(V, \Omega)$ given by

$$(7.1) \quad g((v, a), (w, b)) = \langle v, w \rangle_0 + ab \quad \text{and} \quad \sigma((v, a), (w, b)) = \Omega(v, w)$$

on $\mathfrak{h}(V, \Omega)$. From (4.2), σ is closed; in fact, $\sigma = -d\alpha$, where $\alpha: \mathfrak{h}(V, \Omega) \rightarrow \mathbb{R}$ denotes the second coordinate projection. Note that all left-invariant magnetic systems on $H(V, \Omega)$ having a left-invariant potential 1-form and making V and $\mathbb{R}$ orthogonal are, up to rescaling, of the form (7.1). The Lorentz force operator of

(g, σ) is given by $\mathbf{Y}(v, a) = (\mathbf{Y}_0(v), 0)$, where $\mathbf{Y}_0$ is the operator on V characterized $\langle \mathbf{Y}_0(v), w \rangle_0 = \Omega(v, w)$ for all $v, w \in V$. We similarly have that

$$(7.2) \quad \text{ad}_{(v,a)}(w, b) = (0, \Omega(v, w)) \quad \text{and} \quad \text{ad}_{(v,a)}^\dagger(w, b) = (b\mathbf{Y}_0(v), 0),$$

and hence the only elements (v, a) for which $\text{ad}_{(v,a)}$ is skew-adjoint are precisely the ones with $v = 0$. Setting $\mathbf{z} = (0, 1)$ and identifying (v, a) with $v + a\mathbf{z}$, we use (3.1), (3.2), and (7.2) to obtain

$$(7.3) \quad \nabla_v w = \frac{1}{2}\Omega(v, w)\mathbf{z}, \quad \nabla_{\mathbf{z}} v = \nabla_v \mathbf{z} = -\frac{1}{2}\mathbf{Y}_0(v), \quad \text{and} \quad \nabla_{\mathbf{z}} \mathbf{z} = 0.$$

Noting that $\mathbf{z}$ is central with $\mathbf{Y}(\mathbf{z}) = 0$, and $g([u, v], w) = 0$ for all $u, v, w \in V$, we use (7.3) to evaluate the three coefficients in (2.7), so that

$$(7.4) \quad \begin{aligned} \text{Sec}^{(g, \sigma, s)}(v + a\mathbf{z}, w + b\mathbf{z}) = & s^2 \left(\frac{a^2}{4} \|\mathbf{Y}_0(w)\|^2 + \frac{b^2}{4} \|\mathbf{Y}_0(v)\|^2 \right. \\ & \left. - \frac{3}{4}\Omega(v, w)^2 - \frac{ab}{2} \langle \mathbf{Y}_0(v), \mathbf{Y}_0(w) \rangle \right) \\ & + s \left(\frac{a}{2} \|\mathbf{Y}_0(w)\|^2 - \frac{b}{2} \langle \mathbf{Y}_0(v), \mathbf{Y}_0(w) \rangle \right) \\ & + \frac{3}{4}\Omega(v, w)^2 + \frac{1}{4} \|\mathbf{Y}_0(w)\|^2 \end{aligned}$$

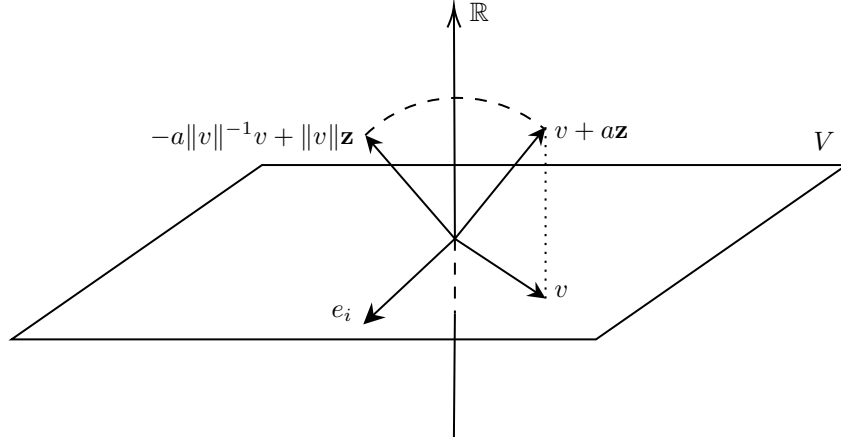
for all orthonormal $v + a\mathbf{z}, w + b\mathbf{z} \in \mathfrak{h}(V, \Omega)$. Notable particular instances of (7.4), for orthonormal $v, w \in V$, are

$$(7.5) \quad \begin{aligned} \text{(i)} \quad & \text{Sec}^{(g, \sigma, s)}(v, w) = \frac{1}{4} (3(1 - s^2)\Omega(v, w)^2 + \|\mathbf{Y}_0(w)\|^2), \\ \text{(ii)} \quad & \text{Sec}^{(g, \sigma, s)}(v, \pm\mathbf{z}) = \frac{s^2}{4} \|\mathbf{Y}_0(v)\|^2, \quad \text{and} \\ \text{(iii)} \quad & \text{Sec}^{(g, \sigma, s)}(\pm\mathbf{z}, w) = \frac{(s \pm 1)^2}{4} \|\mathbf{Y}_0(w)\|^2. \end{aligned}$$

We compute the s -magnetic Ricci curvature of (g, σ) as well, using relation (2.6). Setting $\|\mathbf{Y}_0\| = \sqrt{-\text{tr}(\mathbf{Y}_0^2)} > 0$, it is clear from (7.5-iii) that

$$(7.6) \quad \text{Ric}^{(g, \sigma, s)}(\pm\mathbf{z}) = \frac{(s \pm 1)^2}{4} \|\mathbf{Y}_0\|^2,$$

and it remains to find $\text{Ric}^{(g, \sigma, s)}(v + a\mathbf{z})$, for a unit vector $v + a\mathbf{z} \in \mathfrak{h}(V, \Omega)$ with $v \neq 0$ (which itself does *not* have unit length). One orthonormal basis for the orthogonal complement of $v + a\mathbf{z}$ in $\mathfrak{h}(V, \Omega)$ is $e_1, \dots, e_{n-1}, -a\|v\|_0^{-1}v + \|v\|_0\mathbf{z}$, where $e_1, \dots, e_{n-1}$ is an orthonormal basis for the orthogonal complement of v in V , cf. Figure 1.

FIGURE 1. An orthonormal basis for $(v + a\mathbf{z})^\perp \subseteq \mathfrak{h}(V, \Omega)$.

The unit-length condition $\|v\|^2 + a^2 = 1$ allows for significant simplifications in the expressions to come. Firstly, (7.4) yields

$$(7.7) \quad \text{Sec}^{(\mathbf{g}, \sigma, s)}(v + a\mathbf{z}, -a\|v\|^{-1}v + \|v\|\mathbf{z}) = \frac{(s + a)^2}{4} \frac{\|\mathbf{Y}_0(v)\|^2}{\|v\|^2},$$

while

$$(7.8) \quad \sum_{i=1}^{n-1} \|\mathbf{Y}_0(e_i)\|^2 = \|\mathbf{Y}_0\|^2 - \frac{\|\mathbf{Y}_0(v)\|^2}{\|v\|^2} \quad \text{and} \quad \sum_{i=1}^{n-1} \Omega(v, e_i)^2 = \|\mathbf{Y}_0(v)\|^2$$

now imply that

$$(7.9) \quad \begin{aligned} \sum_{i=1}^{n-1} \text{Sec}^{(\mathbf{g}, \sigma, s)}(v + a\mathbf{z}, e_i) &= \frac{(1 + sa)^2}{4} \left(\|\mathbf{Y}_0\|^2 - \frac{\|\mathbf{Y}_0(v)\|^2}{\|v\|^2} \right) \\ &\quad + \frac{3(1 - s^2)}{4} \|\mathbf{Y}_0(v)\|^2. \end{aligned}$$

Adding (7.7) to (7.9) and factoring further leads to the surprisingly concise formula

$$(7.10) \quad \text{Ric}^{(\mathbf{g}, \sigma, s)}(v + a\mathbf{z}) = \frac{(1 + sa)^2}{4} \|\mathbf{Y}_0\|^2 + \frac{(1 - s^2)}{2} \|\mathbf{Y}_0(v)\|^2.$$

It follows from (7.6) and (7.10) that $\text{Ric}^{(\mathbf{g}, \sigma, s)} > 0$ whenever $0 < s < 1$, while $\text{Ric}^{(\mathbf{g}, \sigma, s_0)}(-\mathbf{z}) = 0$ for $s_0 = 1$ by (7.6), correctly hinting that the Mañé critical value—which marks the most drastic changes in the geometric and dynamical behavior of the corresponding magnetic flow—is $\mathbf{c}(\mathbf{g}, \sigma) = s_0^2/2 = 1/2$, cf. [2, Theorem 2.4] and [25, Theorem 1].

Appendix A. A magnetic Euler-Arnold equation for exact systems

It is well-known [3, 4] that if G is a Lie group and g is a left-invariant Riemannian metric on G , then a curve $\gamma: I \rightarrow G$ is a geodesic if and only if $x(t) = \gamma(t)^{-1}\dot{\gamma}(t)$ satisfies the so-called *Euler-Arnold equation* $\dot{x}(t) = \text{ad}_{x(t)}^\dagger x(t)$. For completeness of our presentation, we exhibit a magnetic generalization of this formalism in Proposition A.2; it has already been considered, for instance, in [33, 38].

We follow the strategy adopted in [18, Proposition 2.1]. The first ingredient needed is what they call the “zero-curvature” equation, and a reference likely to [36, Formula (5.1)]. We instead provide a direct proof.

LEMMA A.1 (The zero-curvature equation). *Given a smooth curve $\gamma: I \rightarrow G$ and a variation $\eta: I \times (-\varepsilon, \varepsilon) \rightarrow G$ of γ , the translated partial derivatives*

$$(A.1) \quad u(t, s) = \eta(t, s)^{-1} \partial_t \eta(t, s) \quad \text{and} \quad v(t, s) = \eta(t, s)^{-1} \partial_s \eta(t, s)$$

satisfy the equation $\partial_s u - \partial_t v = \text{ad}_u v$.

PROOF. Consider the Maurer-Cartan form $\Theta \in \Omega^1(G, \mathfrak{g})$ of G , defined by $\Theta_g(v) = g^{-1}v$, and its well-known structure equation

$$(A.2) \quad d\Theta(X, Y) + [\Theta(X), \Theta(Y)] = 0, \quad \text{for all } X, Y \in \mathfrak{X}(G),$$

cf. [28, Proposition 7.3], which corresponds to flatness of Θ as a connection 1-form on the principal G -bundle $G \rightarrow \{e\}$. Fix also any torsionfree connection ∇ on G , so that the formula

$$(A.3) \quad d\Theta(X, Y) = (\nabla_X \Theta)(Y) - (\nabla_Y \Theta)(X)$$

holds. Finally, noting that $u = \Theta(\partial_t \eta)$ and $v = \Theta(\partial_s \eta)$, we compute:

$$\begin{aligned} (A.4) \quad \partial_s u - \partial_t v &= \partial_s(\Theta(\partial_t \eta)) - \partial_t(\Theta(\partial_s \eta)) \\ &= (\nabla_{\partial_s \eta} \Theta)(\partial_t \eta) + \Theta(\nabla_{\partial_s \eta} \partial_t \eta) - (\nabla_{\partial_t \eta} \Theta)(\partial_s \eta) - \Theta(\nabla_{\partial_t \eta} \partial_s \eta) \\ &= d\Theta(\partial_s \eta, \partial_t \eta) + \Theta(\nabla_{\partial_s \eta} \partial_t \eta - \nabla_{\partial_t \eta} \partial_s \eta) \\ &= -[\Theta(\partial_s \eta), \Theta(\partial_t \eta)] + 0 \\ &= -[v, u] \\ &= \text{ad}_u v, \end{aligned}$$

as required; in the third and fourth lines of (A.4) we use (A.2) and (A.3), respectively, as well as the equality $\nabla_{\partial_s \eta} \partial_t \eta = \nabla_{\partial_t \eta} \partial_s \eta$, valid as ∇ is torsionfree. $\square$

The second ingredient is a variational characterization of magnetic geodesics for exact magnetic systems. Namely, if $(g, -d\alpha)$ is an exact magnetic system on a smooth manifold M , then a smooth curve $\gamma: [a, b] \rightarrow M$ is a $(g, -d\alpha)$ -geodesic

if and only if it is a critical point (under variations with fixed endpoints) of the *Landau-Hall functional*

$$(A.5) \quad E(\gamma) = \int_a^b \left(\frac{1}{2} \|\dot{\gamma}(t)\|^2 + \alpha_{\gamma(t)}(\dot{\gamma}(t)) \right) dt,$$

cf. [13, Section I] (specifically, the Euler-Lagrange equation of (A.5) is precisely (1.1)).

Below, suppose that $(\mathfrak{g}, -d\alpha)$ is a left-invariant magnetic system on the Lie group G , with the potential 1-form α also left-invariant.

PROPOSITION A.2. *A smooth curve $\gamma: [a, b] \rightarrow G$ is a $(\mathfrak{g}, -d\alpha)$ -geodesic if and only if $x(t) = \gamma(t)^{-1}\dot{\gamma}(t)$ satisfies the magnetic Euler-Arnold equation*

$$(A.6) \quad \dot{x}(t) = \text{ad}_{x(t)}^\dagger(x(t) + \alpha^\sharp),$$

where $\alpha^\sharp \in \mathfrak{g}$ is the vector $\mathfrak{g}$ -dual to α . In particular, $(\mathfrak{g}, -d\alpha)$ -geodesics are in one-to-one correspondence with integral curves of the magnetic Euler-Arnold vector field $Z \in \mathfrak{X}(\mathfrak{g})$, defined by $Z_x = \text{ad}_x^\dagger(x + \alpha^\sharp)$ for all $x \in \mathfrak{g}$.

PROOF. Let η be a variation of γ with fixed endpoints, and consider the translated partial derivatives u and v of η as in (A.1). By left-invariance of $\mathfrak{g}$ and α , we may rewrite (A.5) as

$$(A.7) \quad E(\eta(\cdot, s)) = \int_a^b \left(\frac{1}{2} \|u(t, s)\|^2 + \alpha_e(u(t, s)) \right) dt.$$

Differentiating (A.7) with respect to s (omitting arguments and setting $\mathfrak{g} = \langle \cdot, \cdot \rangle$), we obtain

$$(A.8) \quad \begin{aligned} \partial_s E(\eta) &= \int_a^b \langle \partial_s u, u \rangle + \alpha(\partial_s u) dt \\ &\stackrel{(*)}{=} \int_a^b \langle \text{ad}_u v + \partial_t v, u \rangle + \alpha(\text{ad}_u v + \partial_t v) dt \\ &\stackrel{(**)}{=} \int_a^b \langle v, \text{ad}_u^\dagger u - \partial_t u \rangle + \langle \alpha^\sharp, \text{ad}_u v \rangle dt \\ &= \int_a^b \langle v, \text{ad}_u^\dagger u - \partial_t u + \text{ad}_u^\dagger(\alpha^\sharp) \rangle dt. \end{aligned}$$

In $(*)$ we have used the zero-curvature equation, and in $(**)$ integration by parts (noting that all boundary terms vanish as η has fixed endpoints) together with $\alpha = \langle \alpha^\sharp, \cdot \rangle$. Thus γ is a critical point of E if and only if $\text{ad}_x^\dagger x - \dot{x} + \text{ad}_x^\dagger(\alpha^\sharp) = 0$, which is evidently equivalent to (A.6). $\square$

REMARK A.3. The magnetic Euler-Arnold vector field does not depend on the choice of α , in the following sense: if β is another left-invariant 1-form such that

$d\alpha = d\beta$, then $\text{ad}_x^\dagger(x + \alpha^\#) = \text{ad}_x^\dagger(x + \beta^\#)$ for all $x \in \mathfrak{g}$. Indeed, $d(\alpha - \beta) = 0$ means that $(\alpha - \beta)([x, y]) = 0$, and so $\langle \text{ad}_x^\dagger(\alpha^\# - \beta^\#), y \rangle = \langle \alpha^\# - \beta^\#, [x, y] \rangle = 0$ for all $x, y \in \mathfrak{g}$. Hence $\text{ad}_x^\dagger(\alpha^\# - \beta^\#) = 0$, as claimed.

REMARK A.4. Every left-invariant magnetic system is complete. Indeed, (g, σ) is complete whenever g is complete [13, Corollary 2.4] and, more generally, every homogeneous Riemannian metric is complete [31, Corollary 4.4].

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DEPARTMENT OF MATHEMATICS

UNIVERSITY OF CALIFORNIA, RIVERSIDE

RIVERSIDE, CA 92521, USA

Email address: `shane.rankin@email.ucr.edu`

Email address: `ivo.terek@ucr.edu`

Email address: `david.weed@email.ucr.edu`